

Continuum Mechanics
class # 05
9/2/2026

Fourth-Order Dyadic Products

Definition: Suppose that $\vec{a}, \vec{b}, \vec{c}, \vec{d} \in V$. The object

$$\vec{a} \otimes \vec{b} \otimes \vec{c} \otimes \vec{d}$$

is a fourth-order tensor called the dyadic product and is defined via the action

$$(\vec{a} \otimes \vec{b} \otimes \vec{c} \otimes \vec{d})_T = (\vec{c} \cdot T \vec{d}) \vec{a} \otimes \vec{b}$$

for all $T \in V^2$.

Proposition: Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame for V . The set

$$\{\vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l \mid 1 \leq i, j, k, l \leq 3\}$$

is a basis for V^4 . In particular, for any $C \in V^4$,

$$(1.19) \quad C = C_{ijkl} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l$$

where

$$C_{ijkl} = \vec{e}_i \cdot (C(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j)$$

Proof: Let $C \in V^4$ be given. Suppose $U, T \in V^2$ satisfy

$$U = CT.$$

Suppose that

$$U = U_{ij} \vec{e}_i \otimes \vec{e}_j$$

and

$$T = T_{ij} \vec{e}_i \otimes \vec{e}_j$$

Recall from (1.18) that

$$U_{ij} = C_{ijkl} T_{kl} \Leftrightarrow V = CT$$

By the defn of new dyadic product

$$\begin{aligned} C_{ijkl} (\vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l) T \\ &= C_{ijkl} (\vec{e}_k \cdot T \vec{e}_l) \vec{e}_i \otimes \vec{e}_j \\ &= C_{ijkl} T_{kl} \vec{e}_i \otimes \vec{e}_j \\ &= U_{ij} \vec{e}_i \otimes \vec{e}_j \\ &= U \end{aligned}$$

Thus, for any $T \in V^2$,

$$C_{ijkl} (\vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l) T = CT,$$

which implies that

$$C_{ijkl} (\vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l) = C. //$$

Symmetry Properties

Definition: The fourth-order tensor $C \in V^4$ is said to have major symmetry iff

$$A : C(B) = C(A) : B, \quad \forall A, B \in V^2.$$

It is said to have right minor symmetry iff

$$A : C(B) = A : C(\text{sym}(B)), \quad \forall A, B \in V^2$$

It is said to have left minor symmetry iff

$$A : C(B) = \text{sym}(A) : C(B), \quad \forall A, B \in V^2$$

Result (1.14): If $C \in V^4$ has major symmetry, then

$$C_{ijkl} = C_{klij}, \quad \forall i, j, k, l \in \{1, 2, 3\}.$$

If C has right and left minor symmetries, then

$$C_{ijkl} = C_{ijlk} \quad \text{and} \quad C_{ijkl} = C_{jilk},$$

respectively.

Proof: Suppose that $C \in V^4$ has major symmetry. Then,

$$\begin{aligned} C_{ijkl} &= \vec{e}_i \cdot C(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j \\ &= \vec{e}_i \otimes \vec{e}_j : C(\vec{e}_k \otimes \vec{e}_l) \\ &= C(\vec{e}_i \otimes \vec{e}_j) : \vec{e}_k \otimes \vec{e}_l \quad (\text{major sym.}) \\ &= \vec{e}_k \otimes \vec{e}_l : C(\vec{e}_i \otimes \vec{e}_j) \quad (\text{sym of :}) \\ &= C_{klij} \quad \parallel \end{aligned}$$

The other computations are left as exercises. $\parallel$

Definition: A function $G: \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is called isotropic iff

$$QG(A)Q^T = G(QAQ^T)$$

for any $A \in \mathcal{V}^2$ and for all rotations Q .

A function $g: \mathcal{V}^2 \rightarrow \mathbb{R}$ is called isotropic iff

$$g(A) = g(QAQ^T)$$

for any $A \in \mathcal{V}^2$ and for all rotations Q .

Result (1.15): let $G: \mathcal{V}^2 \rightarrow \mathcal{V}^2$ be an isotropic function with the property that, if $A \in \mathcal{V}^2$ is symmetric then $G(A)$ is symmetric. Then there are functions $\alpha_0, \alpha_1, \alpha_2: \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$G(A) = \alpha_0(\vec{I}_A) I + \alpha_1(\vec{I}_A) A + \alpha_2(\vec{I}_A) A^2,$$

for every symmetric tensor $A \in \mathcal{V}^2$.

Here $\vec{I}_A = (I_1(A), I_2(A), I_3(A))$, the 3 invariants of A .

By Result 1.10, there are functions $\beta_0, \beta_1, \beta_2: \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$G(A) = \beta_0(\vec{I}_A) I + \beta_1(\vec{I}_A) A + \beta_2(\vec{I}_A) A^{-1},$$

for every symmetric and invertible tensor $A \in \mathcal{V}^2$.

Proof: Gurtin, M.E., An Intro. to Contin. Mechanics, Academic Press (1981). //

Remark: Result 1.10 is the Cayley-Hamilton Theorem.

Result (1.16): let $C \in \mathcal{V}^4$ be an isotropic fourth-order tensor with the properties

(1) $C(A) \in \mathcal{V}^2$ is symmetric for every symmetric $A \in \mathcal{V}^2$.

(2) $C(W) = 0$ for every skew symmetric $W \in \mathcal{V}^2$.

Then there are scalars μ and λ such that

$$C(A) = \lambda \operatorname{tr}(A) \mathbf{I} + 2\mu \operatorname{Sym}(A),$$

for all $A \in \mathcal{V}^2$.

Proof: Since $C \in \mathcal{V}^4$ is isotropic and maps symmetric tensors to symmetric tensors, it follows from Result 1.15 that there are functions $\alpha_0, \alpha_1, \alpha_2: \mathbb{R}^3 \rightarrow \mathbb{R}$, such that

$$C(H) = \alpha_0(\vec{\mathcal{I}}_H) \mathbf{I} + \alpha_1(\vec{\mathcal{I}}_H) H + \alpha_2(\vec{\mathcal{I}}_H) H^2$$

for every symmetric $H \in \mathcal{V}^2$. Here

$$\vec{\mathcal{I}}_H = (\operatorname{tr}(H), \frac{1}{2}[(\operatorname{tr} H)^2 - \operatorname{tr}(H^2)], \det H)$$

Since C is linear in H ,

$$\alpha_0(\vec{\mathcal{I}}_H) = c_0 \operatorname{tr}(H) + c_1,$$

$$\alpha_1(\vec{\mathcal{I}}_H) = c_2$$

$$\alpha_2(\vec{\mathcal{I}}_H) = 0$$

where $c_0, c_1, c_2 \in \mathbb{R}$ are constants.

Since $C(O) = O$ (because O is ^{symmetric and} skew symmetric), it must be that

$$O = C(O) = (c_0 \text{tr}(O) + c_1)I + c_2 O \\ = c_1 I$$

Thus, $c_1 = 0$. Now set $\lambda = c_0$ and $\mu = c_2/2$.
Then

$$C(H) = \lambda \text{tr}(H)I + 2\mu H$$

for every symmetric matrix H .

Let $A \in V^2$ be arbitrary. Then

$$A = \text{Sym}(A) + \text{Skew}(A)$$

Since C is linear

$$C(A) = C(\text{Sym}(A)) + C(\text{Skew}(A)) \\ = \lambda \text{tr}(\text{Sym}(A))I + 2\mu \text{Sym}(A) \\ = \lambda \text{tr}(A)I + 2\mu \text{Sym}(A). \quad |||$$

Example: (Exercise 1.7): Given $\vec{a}, \vec{b} \in V$ and $S \in V^2$,
prove

$$a) S(\vec{a} \otimes \vec{b}) = (S\vec{a}) \otimes \vec{b}$$

$$b) (\vec{a} \otimes \vec{b})S = \vec{a} \otimes (S^T \vec{b})$$

$$(c) (\vec{a} \otimes \vec{b})^T = (\vec{b} \otimes \vec{a}).$$

(a): let $\vec{v} \in V$ be arbitrary. Then

$$\begin{aligned} (S(\vec{a} \otimes \vec{b}))\vec{v} &= S((\vec{a} \otimes \vec{b})\vec{v}) \\ &= S((\vec{b} \cdot \vec{v})\vec{a}) \\ &= (\vec{b} \cdot \vec{v}) S\vec{a} \\ &= (S\vec{a} \otimes \vec{b})\vec{v} \end{aligned}$$

Thus

$$S(\vec{a} \otimes \vec{b}) = (S\vec{a}) \otimes \vec{b},$$

(b): let $\vec{v} \in V$ be arbitrary. Then

$$\begin{aligned} ((\vec{a} \otimes \vec{b})S)\vec{v} &= (\vec{a} \otimes \vec{b})(S\vec{v}) \\ &= (\vec{b} \cdot (S\vec{v}))\vec{a} \\ &= (\vec{v} \cdot (S^T\vec{b}))\vec{a} \\ &= ((S^T\vec{b}) \cdot \vec{v})\vec{a} \\ &= (\vec{a} \otimes (S^T\vec{b}))\vec{v}. \end{aligned}$$

Since $\vec{v}$ is arbitrary,

$$(\vec{a} \otimes \vec{b})S = \vec{a} \otimes (S^T\vec{b}),$$

(c): let $\vec{u}, \vec{v} \in V$ be arbitrary. Then

$$\begin{aligned}
 \vec{w} \cdot ((\vec{a} \otimes \vec{b})^T \vec{v}) &= \vec{v} \cdot ((\vec{a} \otimes \vec{b}) \vec{w}) \\
 &= \vec{v} \cdot ((\vec{b} \cdot \vec{w}) \vec{a}) \\
 &= (\vec{v} \cdot \vec{a})(\vec{b} \cdot \vec{w}).
 \end{aligned}$$

Suppose $\vec{v} = \vec{e}_j$ $\vec{w} = \vec{e}_i$. Then

$$\begin{aligned}
 [(\vec{a} \otimes \vec{b})^T]_{ij} &= b_i a_j \\
 &= [\vec{b} \otimes \vec{a}]_{ij}.
 \end{aligned}$$

Thus

$$(\vec{a} \otimes \vec{b})^T = \vec{b} \otimes \vec{a} \quad ///$$

Example: (Exercise 1.24): Let $S \in V^2$. Suppose that

$$I_2 = \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)].$$

Show that I_2 has the same numerical value regardless of the frame in which it is calculated.

Solution: Suppose $\mathcal{F} = \{\vec{e}_i\}$ and $\mathcal{F}' = \{\vec{e}'_i\}$ are two frames and $[S]$ and $[S]'$ are the matrix representations in those frames, respectively. We have already seen (in Result 1.5) that

$$\text{tr}[S] = \text{tr}[S'].$$

Now, let $[S^2]$ and $[S^2]'$ be representations in the respective frames. Let A be the change of basis tensor from $\mathcal{F}$ to $\mathcal{F}'$. Then

$$[S^2] = [A][S^2]'[A]^T$$

and

$$[S] = [A][S]'[A]^T$$

Therefore,

$$\text{tr}[S^2] = \text{tr}[S^2]'$$

and

$$\begin{aligned} I_2(S) &= \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)] \\ &= \frac{1}{2} [(\text{tr}[S])^2 - \text{tr}[S^2]] \\ &= \frac{1}{2} [(\text{tr}[S]')^2 - \text{tr}[S^2]']. \quad /// \end{aligned}$$

Example: (Exercise (1.32)): Find the components of the tensor $C \in V^4$ defined by

$$C(S) = \frac{1}{2}(S - S^T)$$

Solution: By definition

$$\begin{aligned} C_{ijkl} &= \vec{e}_i \cdot (C(\vec{e}_k \otimes \vec{e}_l) \vec{e}_j) \\ &= \vec{e}_i \cdot \left[\left(\frac{1}{2} \{ (\vec{e}_k \otimes \vec{e}_l) - (\vec{e}_k \otimes \vec{e}_l)^T \} \right) \vec{e}_j \right] \\ &= \vec{e}_i \cdot \left[\left(\frac{1}{2} \{ (\vec{e}_k \otimes \vec{e}_l) - (\vec{e}_l \otimes \vec{e}_k) \} \right) \vec{e}_j \right] \\ &= \vec{e}_i \cdot \left[\frac{1}{2} (\vec{e}_k \otimes \vec{e}_l) \vec{e}_j - \frac{1}{2} (\vec{e}_l \otimes \vec{e}_k) \vec{e}_j \right] \\ &= \frac{1}{2} \vec{e}_i \cdot [(\vec{e}_l \cdot \vec{e}_j) \vec{e}_k - (\vec{e}_k \cdot \vec{e}_j) \vec{e}_l] \\ &= \frac{1}{2} \vec{e}_i \cdot [\delta_{lj} \vec{e}_k - \delta_{kj} \vec{e}_l] \\ &= \frac{1}{2} \delta_{ik} \delta_{lj} - \frac{1}{2} \delta_{il} \delta_{kj} \quad /// \end{aligned}$$